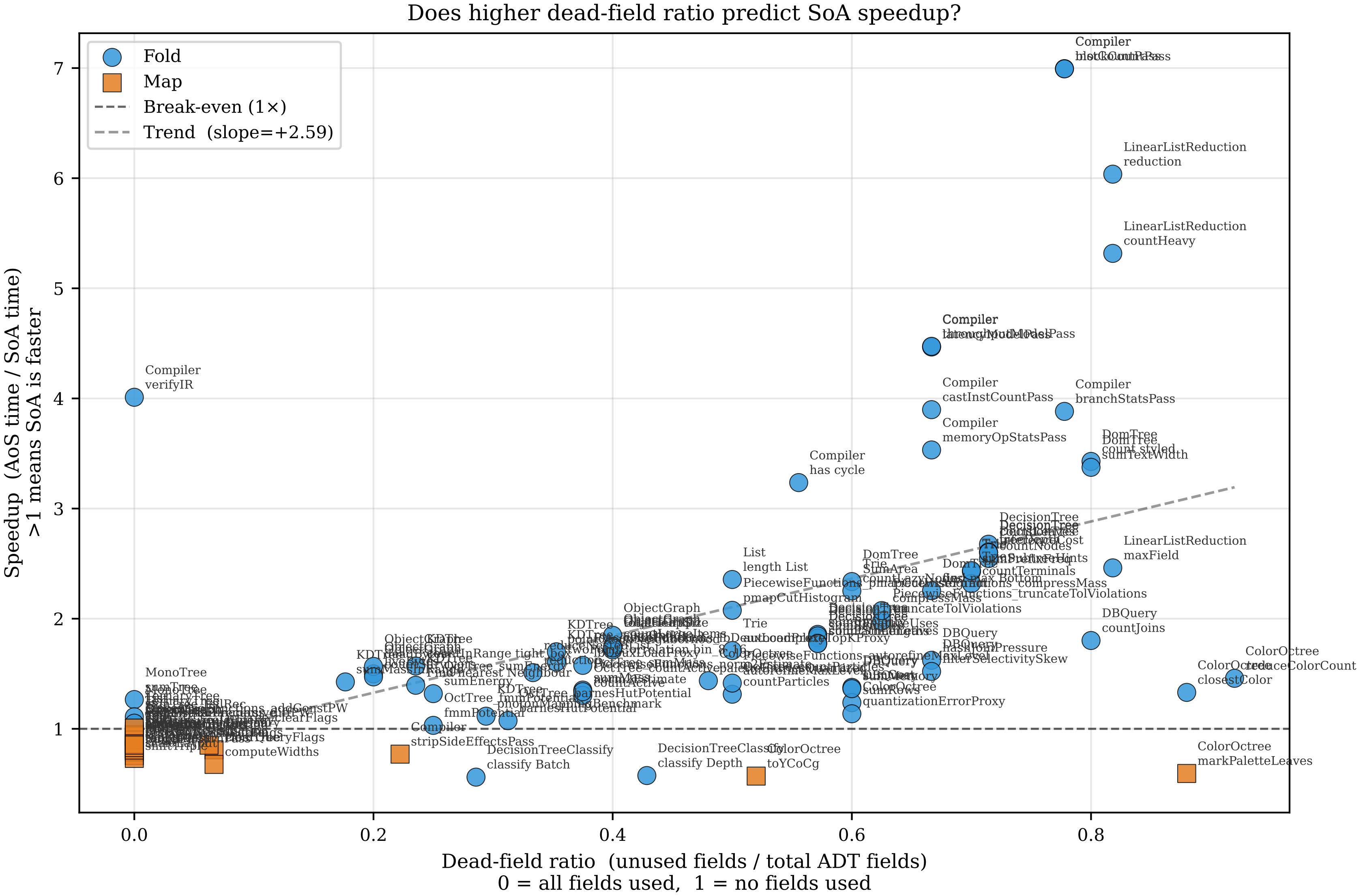
Does higher dead-field ratio predict SoA speedup?
Fold
Map
Break-even (1×)
Trend (slope=+2.59)
Speedup (AoS time / SoA time)
>1 means SoA is faster
Dead-field ratio (unused fields / total ADT fields)
0 = all fields used, 1 = no fields used
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8
Compiler
LinearListReduction
reduction
LinearListReduction
countHeavy
Compiler
Compiler
castInstCountPass
Compiler
branchStatsPass
Compiler
memoryOpStatsPass
Compiler
has cycle
Compiler
verifyIR
DomTree
DomTree
sumTextWidth
DecisionTree
LinearListReduction
maxField
List
length List
PiecewiseFunctions_pmapCutHistogram
DomTree
Trie
sumArea
DomTree
countTerminals
PiecewiseFunctions_compressMass
PiecewiseFunctions_truncateTolViolations
compressMass
truncateTolViolations
DBQuery
countJoins
DBQuery
DBQuery
filterSelectivitySkew
ObjectGraph
KDTree
Trie
ColorOctree
reduceColorCount
ColorOctree
closestColor
ColorOctree
sumRows
quantizationErrorProxy
countParticles
MonoTree
sumTree
sumEnergy
OctTree_fmmPotential
fmmPotential
KDTree
photonMappingBenchmark
barnesHutPotential
OctTree_barnesHutPotential
countActive
Compiler
stripSideEffectsPass
computeWidths
DecisionTreeClassify
classify Batch
DecisionTreeClassify
classify Depth
ColorOctree
toYCoCg
ColorOctree
markPaletteLeaves